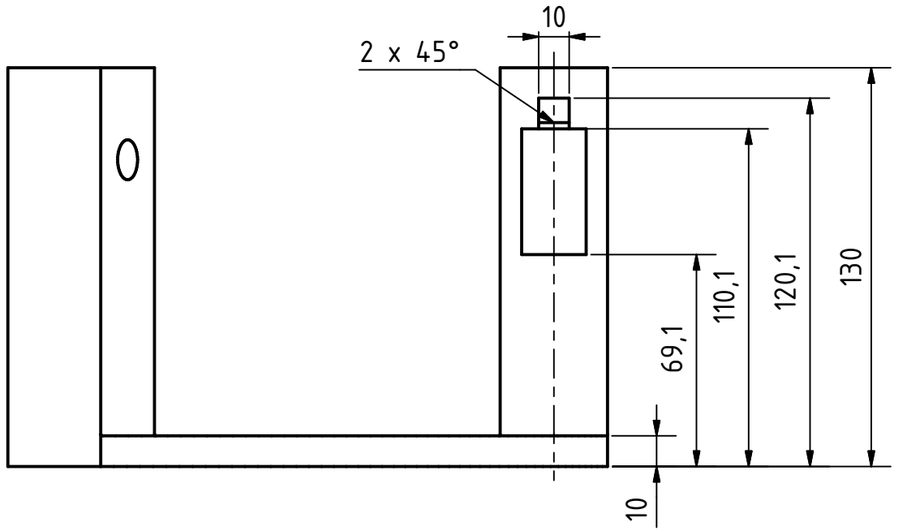
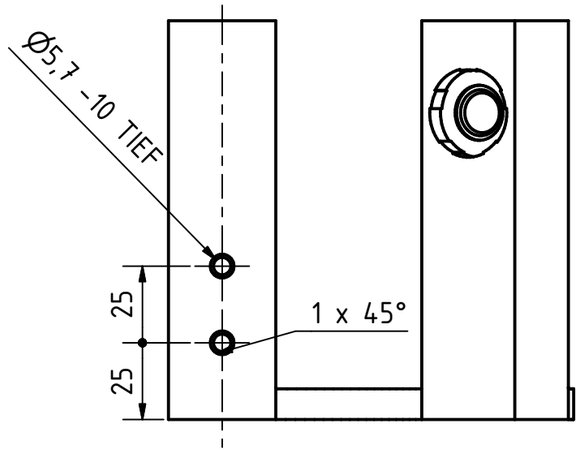
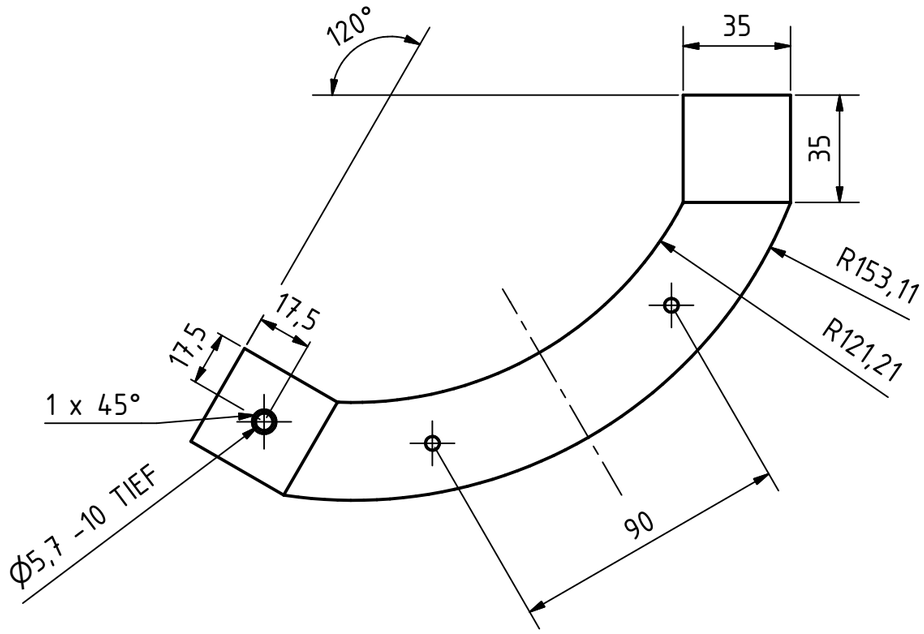
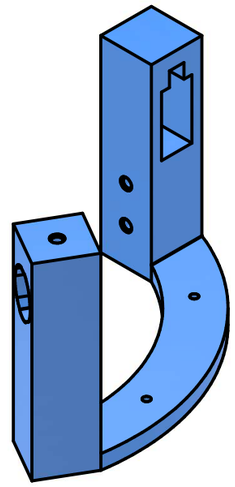








ingenieur
wissenschaften
htw saar

					Datum
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Maßstab: 1:2	

Industrielle Robotik
Ball Ballancer

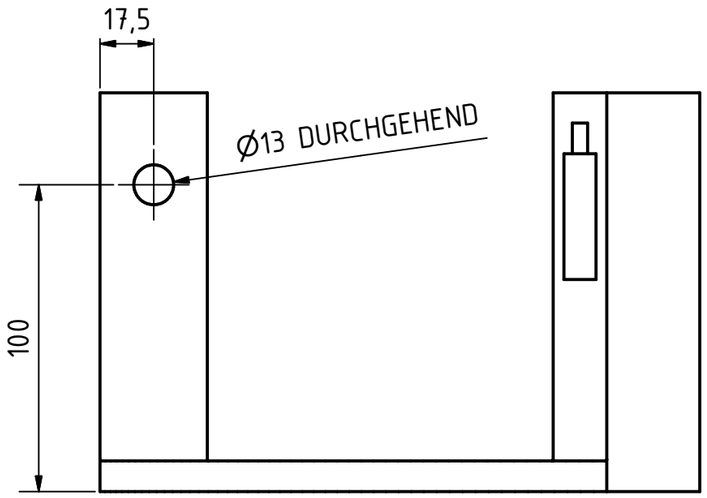
Pos. Nr. 1

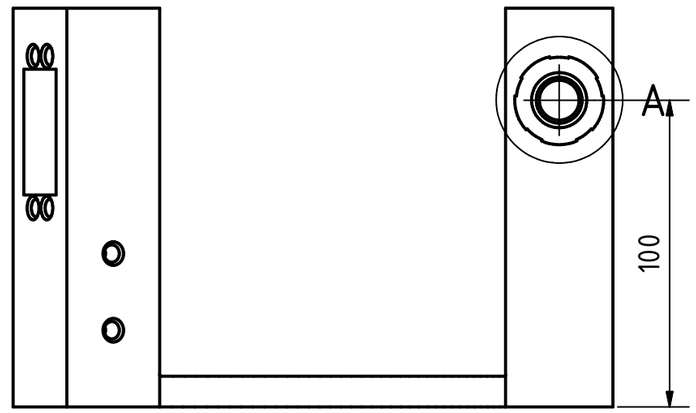
Grundhalter

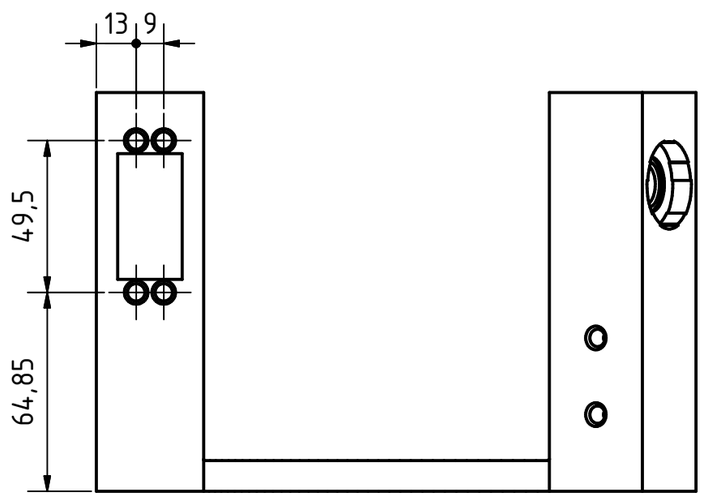
1

A3

Konstrukteur: Simon Bernarding







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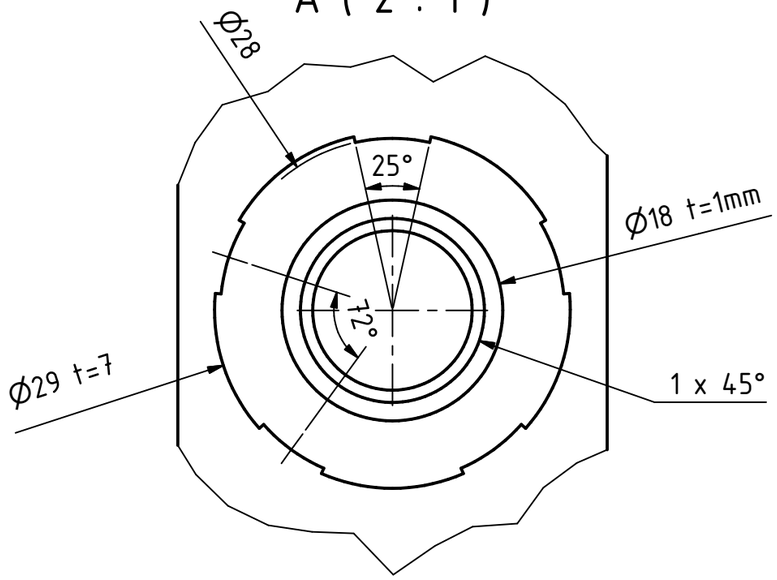
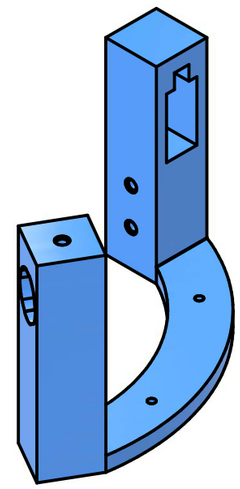




ingenieur
wissenschaften
htw saar

				Datum	
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Maßstab: 1:2	

Industrielle Robotik
Ball Ballancer

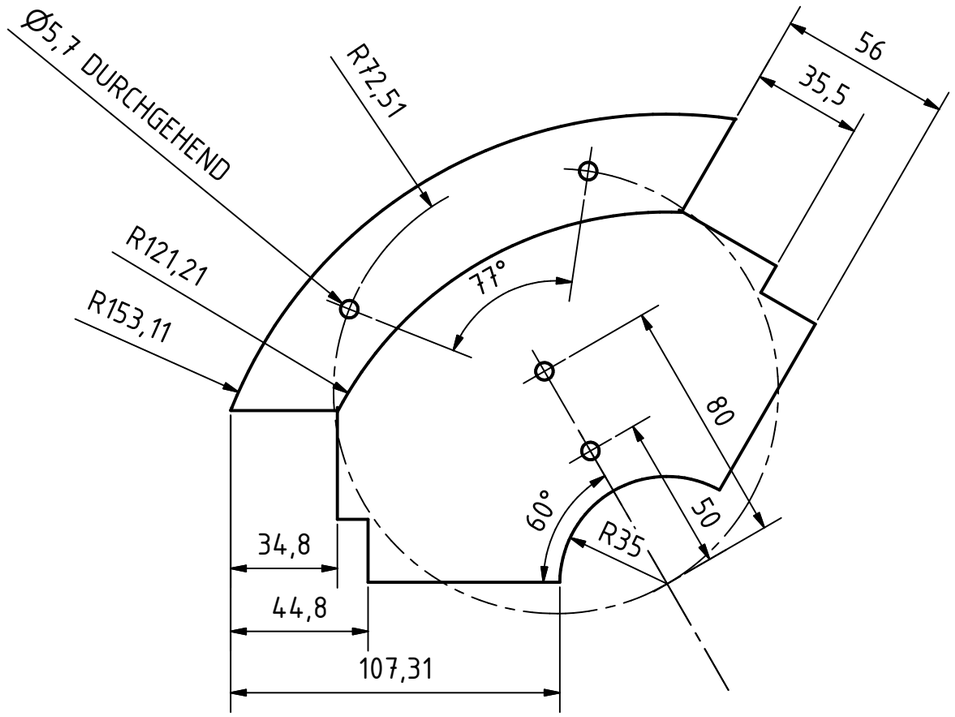
Pos. Nr. 1

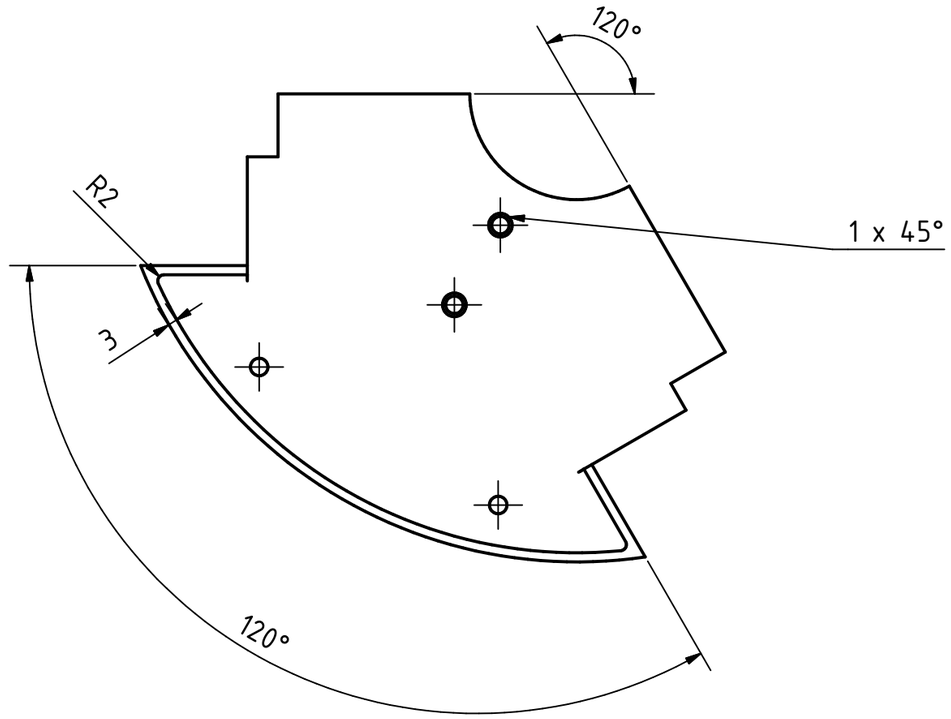
Grundhalter

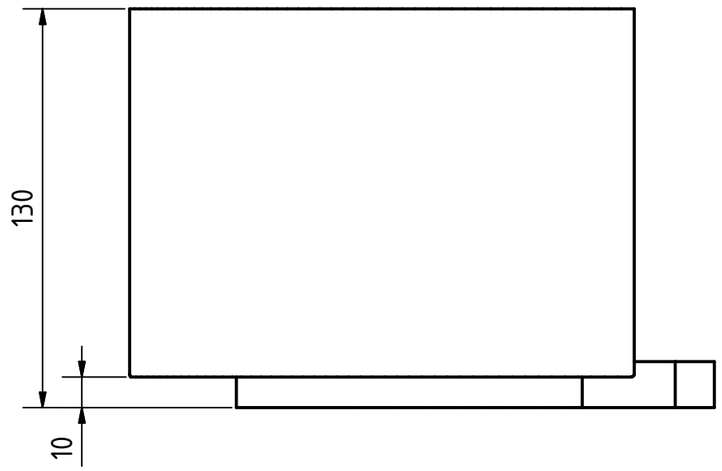
2

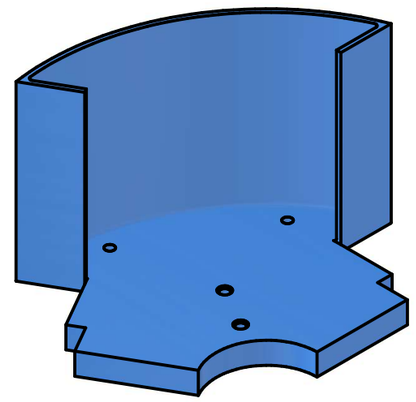
A3

Konstrukteur: Simon Bernarding









ingenieur
wissenschaften
htw saar

			Datum
		Gezeich	25.03.2026
Status	Änderungen	Datum	Name

Industrielle Robotik
Ball Ballancer

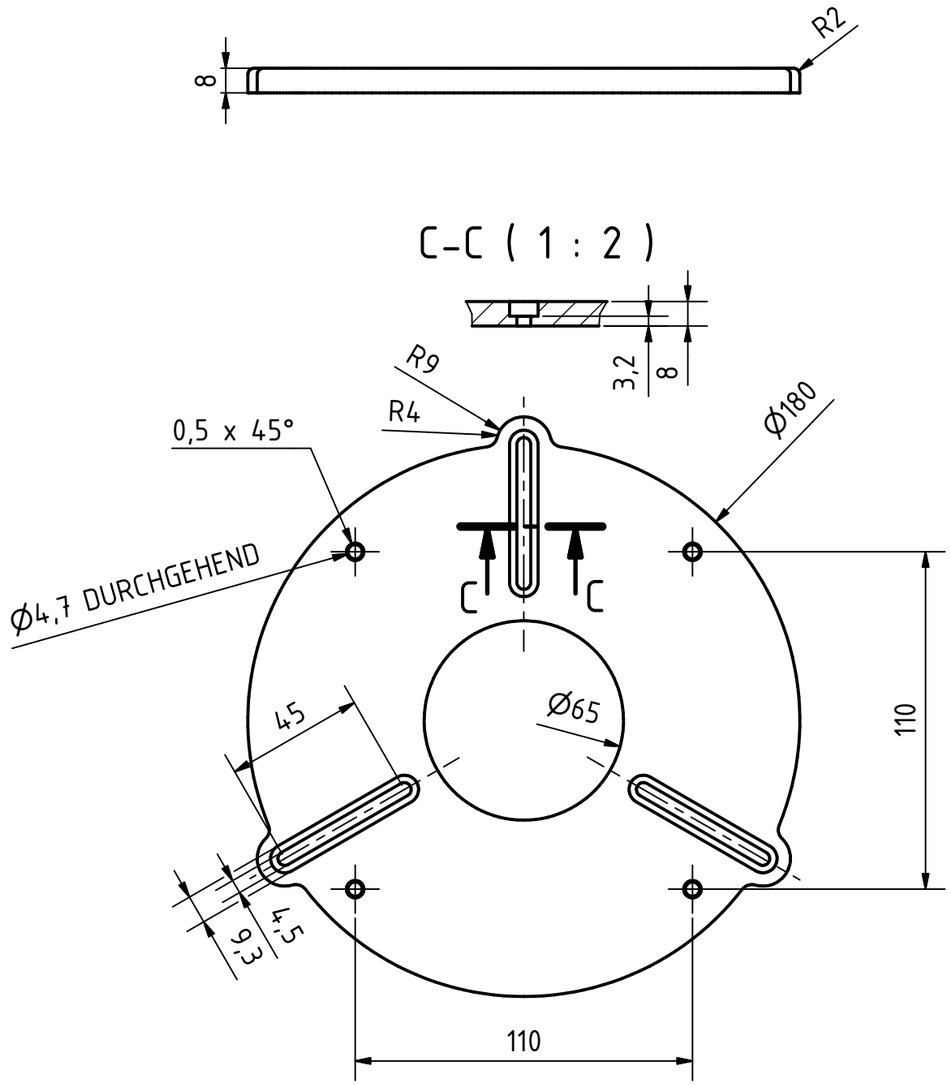
Pos. Nr. 2

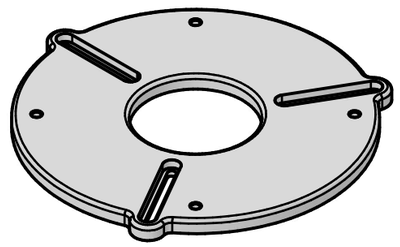
Abdeckung

3
A3

Maßstab: 1:2

Konstrukteur: Simon Bernarding





ingenieur
wissenschaften
htw saar

					Datum
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Maßstab: 1:2	

Industrielle Robotik
Ball Ballancer

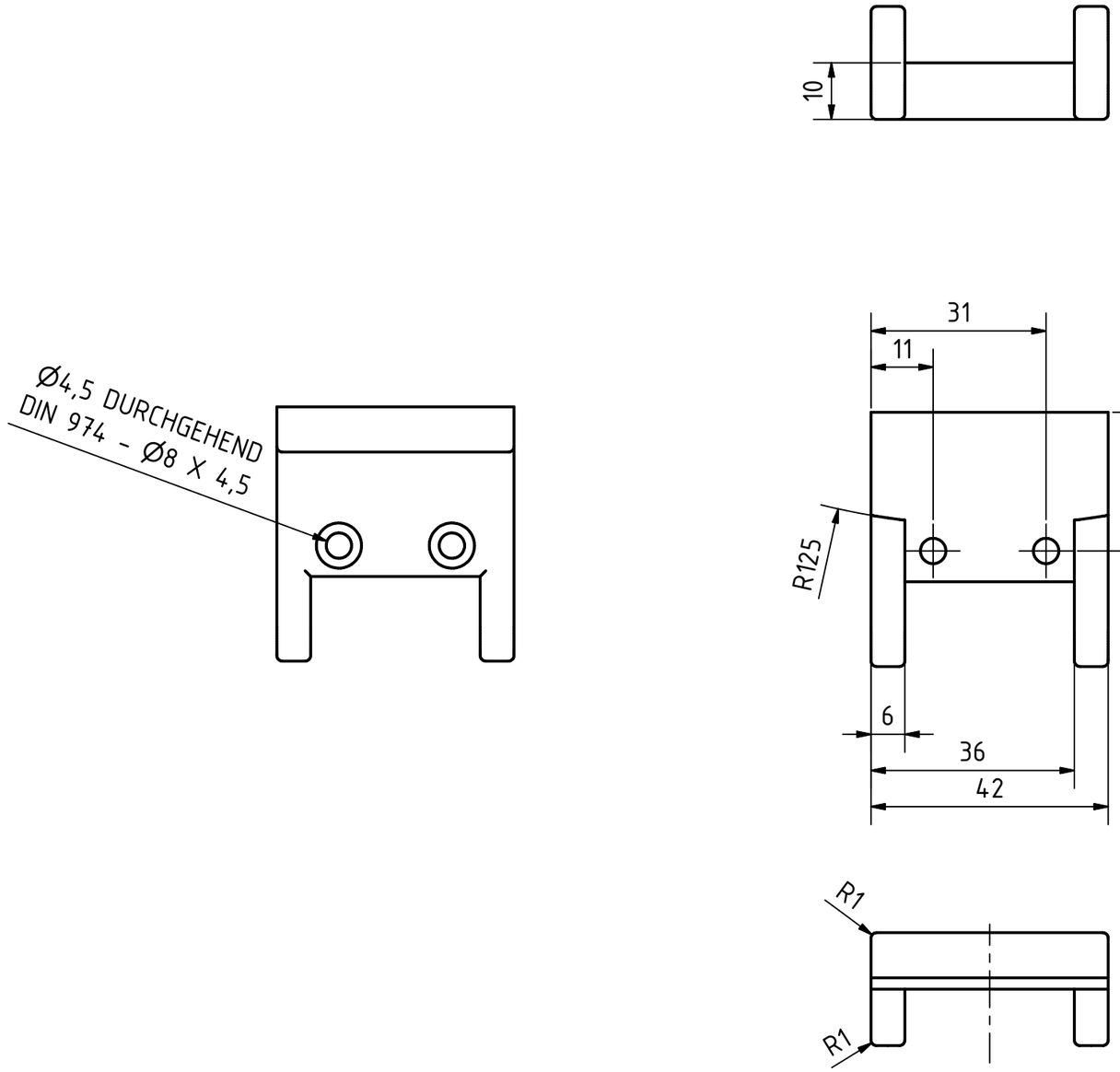
Pos. Nr. 3

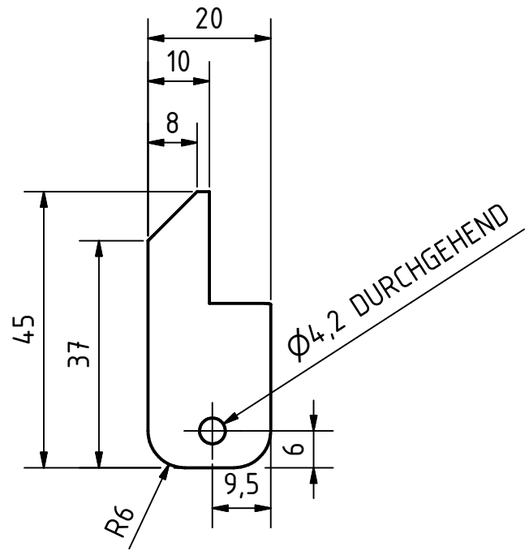
Innere Grundplatte

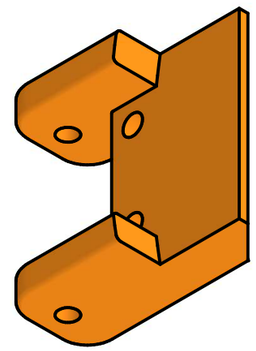
4

A3

Konstrukteur: Simon Bernarding







ingenieur
wissenschaften
htw saar

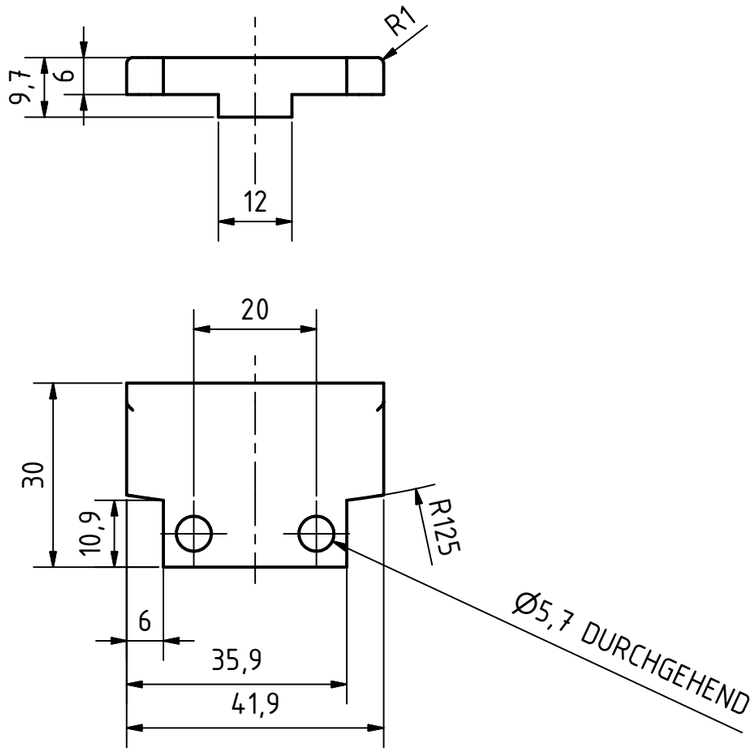
				Datum	
				Gezeich	25.03.2026
				Pos. Nr. 4	
Status	Änderungen	Datum	Name	Maßstab: 1:1	

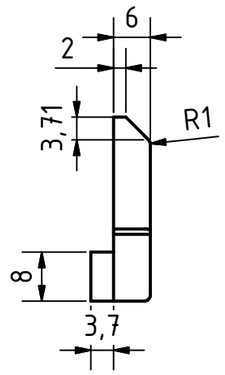
Industrielle Robotik
Ball Ballancer

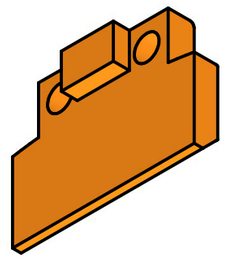
Klemme Unten

5
A3

Konstrukteur: Simon Bernarding







ingenieur
wissenschaften
htw saar

					Datum
				Gezeich	25.03.2026
				Pos. Nr. 5	
Status	Änderungen	Datum	Name	Maßstab: 1:1	

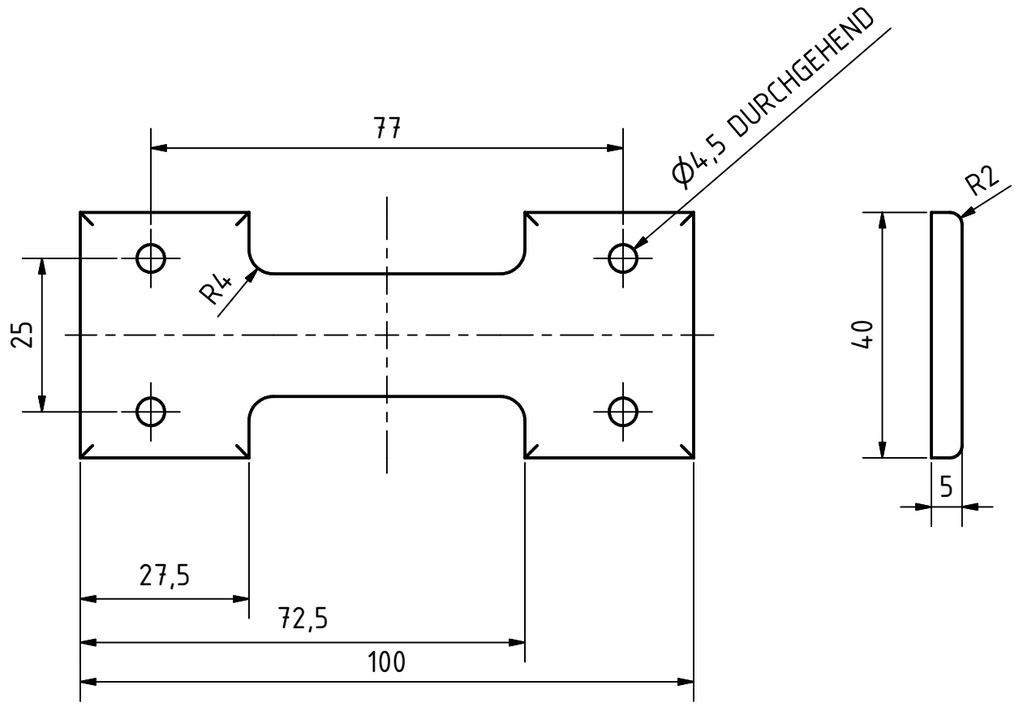
Industrielle Robotik
Ball Ballancer

Klemme Oben

6

A3

Konstrukteur: Simon Bernarding



ingenieur
wissenschaften
htw saar

					Datum
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Maßstab: 1:1	

Industrielle Robotik
Ball Ballancer

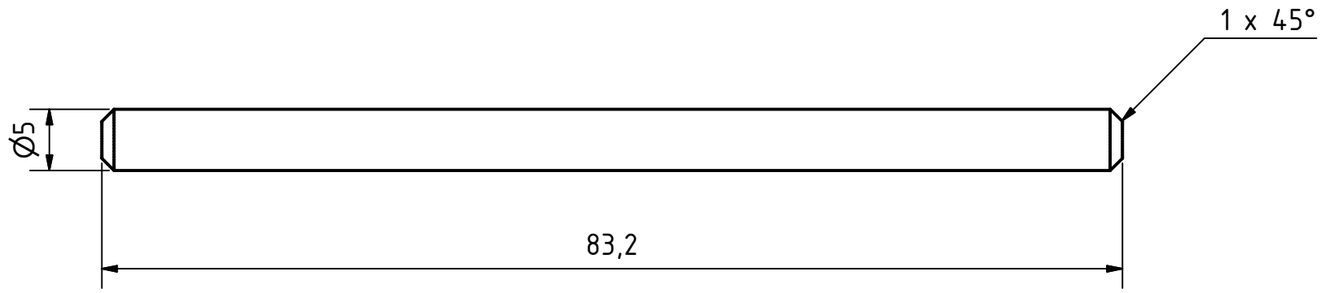
Pos. Nr. 6

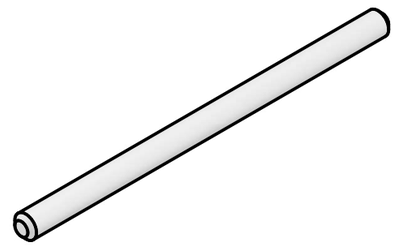
Verspannung Lagerturm

7

A3

Konstrukteur: Simon Bernarding





ingenieur
wissenschaften
htw saar

					Datum
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Maßstab: 1:1	

Industrielle Robotik
Ball Ballancer

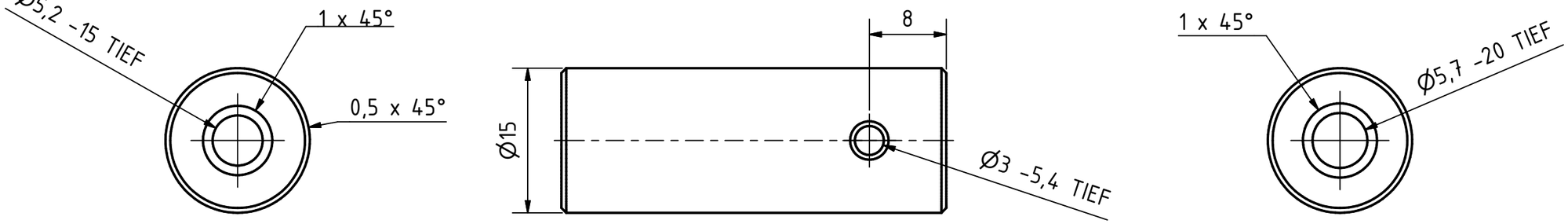
Pos. Nr. 7

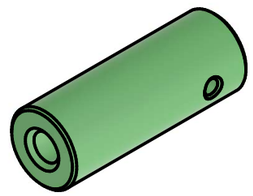
Verbinder Schwinge

8

A3

Konstrukteur: Simon Bernarding





ingenieur
wissenschaften
htw saar

					Datum
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Maßstab: 2:1	

Industrielle Robotik
Ball Ballancer

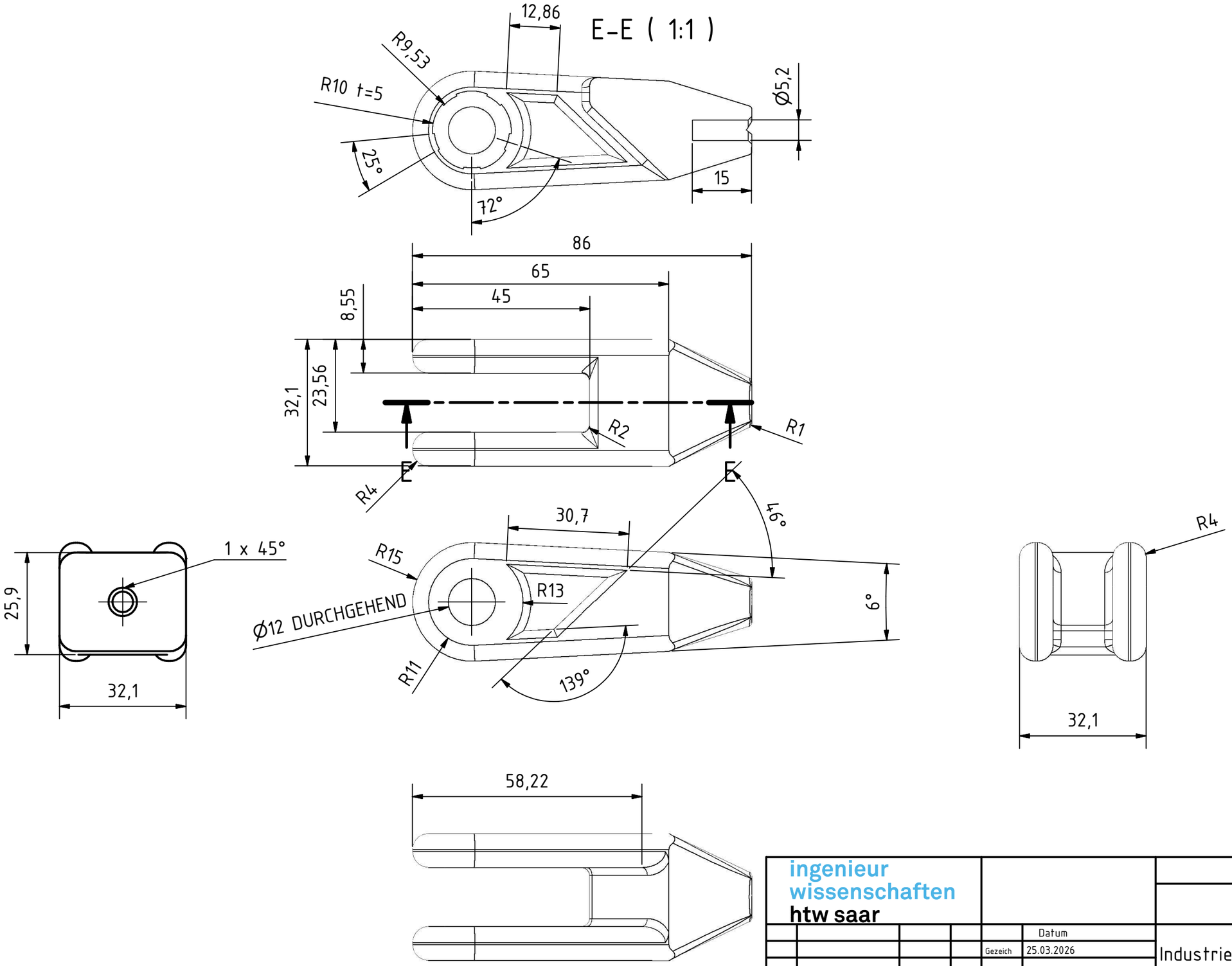
Pos. Nr. 8

Schwinge Oben

9

A3

Konstrukteur: Simon Bernarding



ingenieur
wissenschaften
htw saar

		Datum	
		Gezeich	25.03.2026
Status	Änderungen	Datum	Name

Industrielle Robotik
Ball Ballancer

Pos. Nr. 9

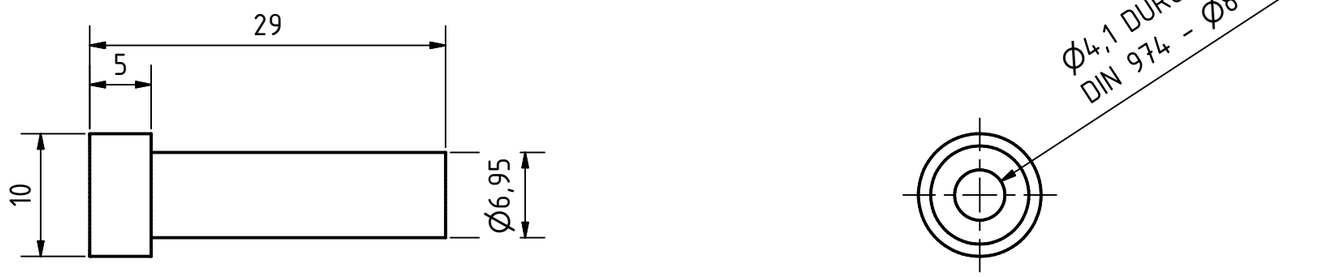
Schwinge Unten

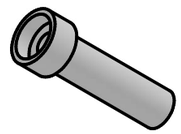
10

A3

Maßstab: 1:1

Konstrukteur: Simon Bernarding





ingenieur
wissenschaften
htw saar

					Datum
				Gezeich	25.03.2026
				Pos. Nr. 10	
Status	Änderungen	Datum	Name	Maßstab: 2:1	

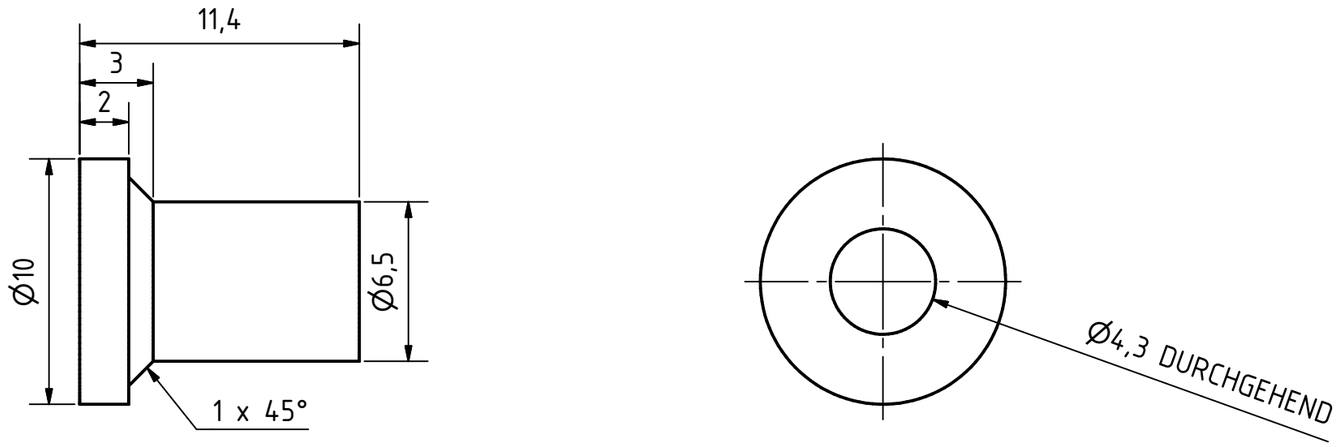
Industrielle Robotik
Ball Ballancer

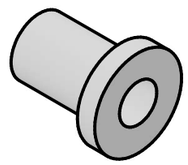
Lager Bolzen Knickarm

11

A3

Konstrukteur: Simon Bernarding





ingenieur
wissenschaften
htw saar

					Datum
				Gezeich	25.03.2026
				Pos. Nr. 11	
Status	Änderungen	Datum	Name	Maßstab: 4:1	

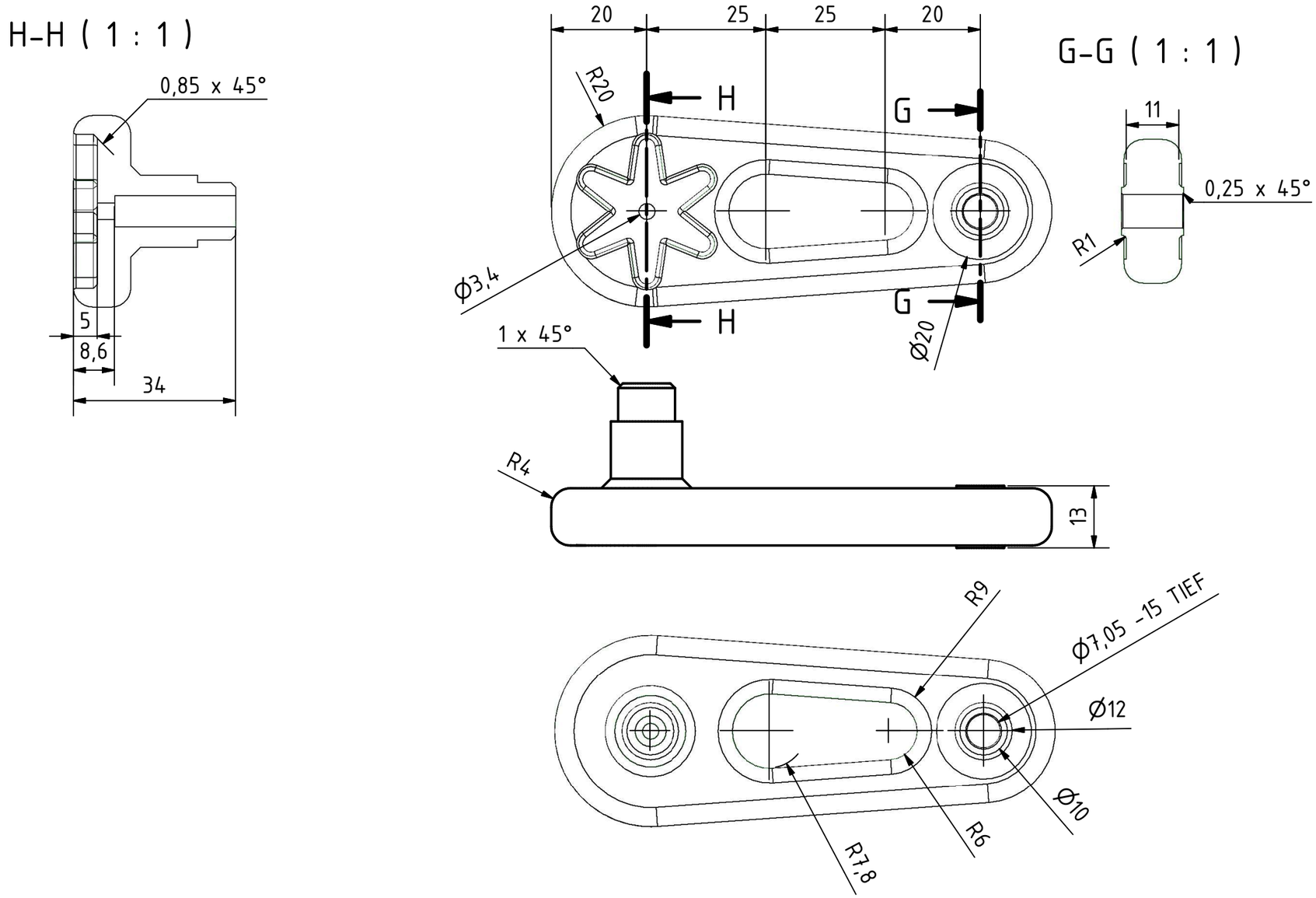
Industrielle Robotik
Ball Ballancer

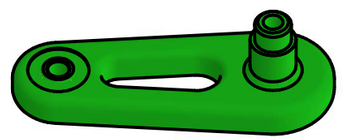
Abstandhalter Klemme

12

A3

Konstrukteur: Simon Bernarding





ingenieur
wissenschaften
htw saar

		Datum	
		Gezeich	25.03.2026
Status	Änderungen	Datum	Name

Industrielle Robotik
Ball Ballancer

Pos. Nr. 12

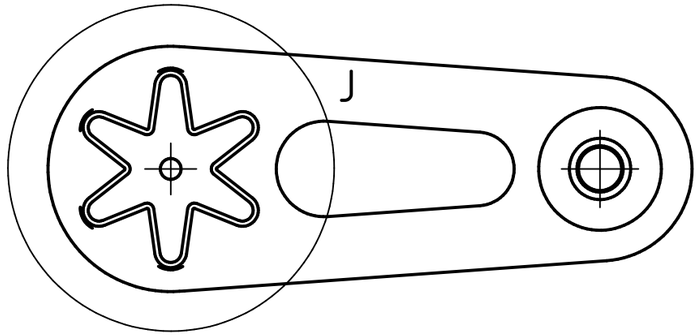
Servo Kurbel

13

A3

Maßstab: 1:1

Konstrukteur: Simon Bernarding



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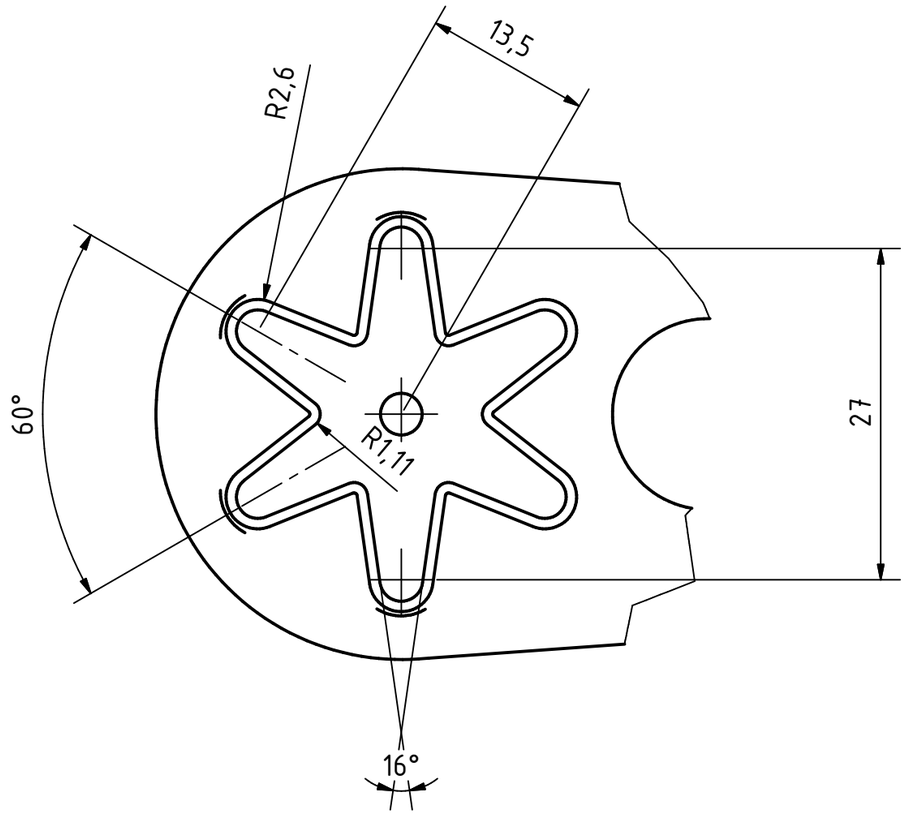


ingenieur
wissenschaften
htw saar

				Datum	
				Gezeich	25.03.2026
				Pos. Nr. 12	
Status	Änderungen	Datum	Name	Maßstab: 1:1	

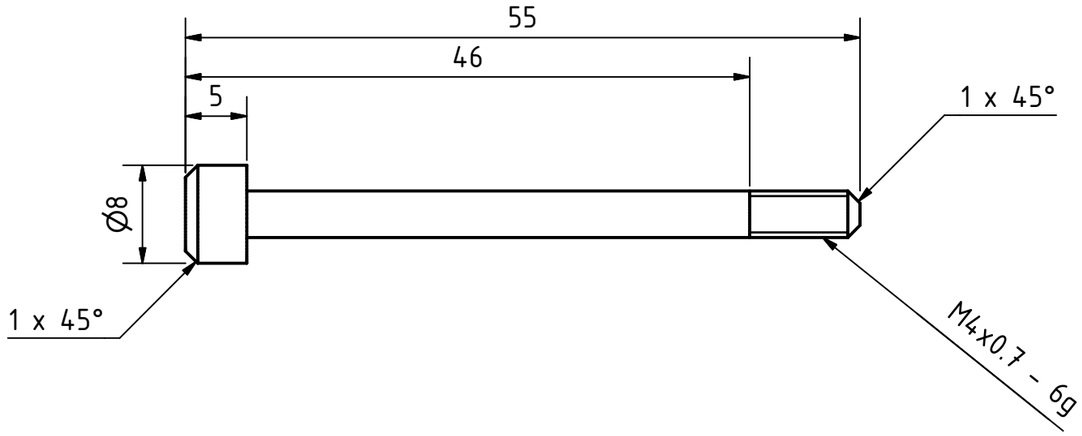
Industrielle Robotik
Ball Ballancer

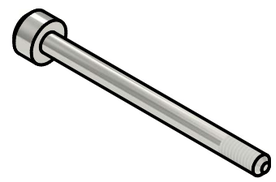
Servo Kurbel

14

A3

Konstrukteur: Simon Bernarding





ingenieur
wissenschaften
htw saar

					Datum
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Maßstab: 2:1	

Industrielle Robotik
Ball Ballancer

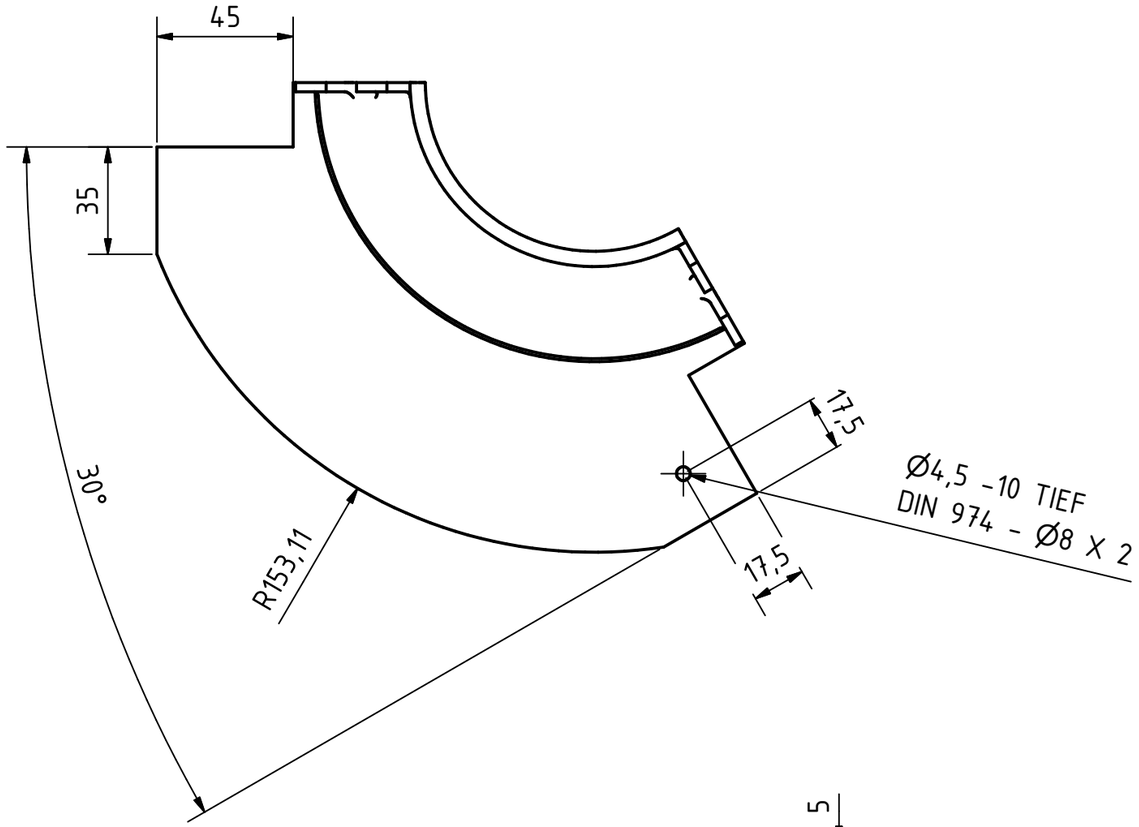
Pos. Nr. 13

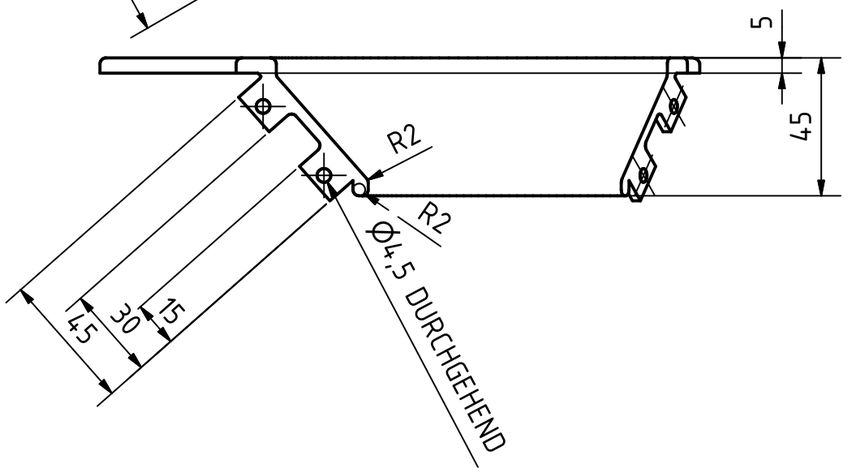
Bolzen Klemme

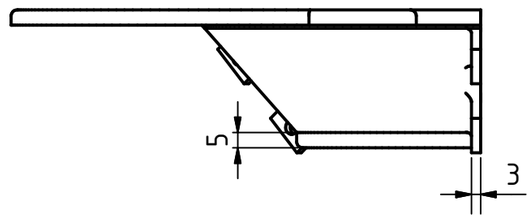
15

A3

Konstrukteur: Simon Bernarding







ingenieur
wissenschaften
htw saar

		Datum	
		Gezeich	25.03.2026
Status	Änderungen	Datum	Name

Industrielle Robotik
Ball Ballancer

Pos. Nr. 14

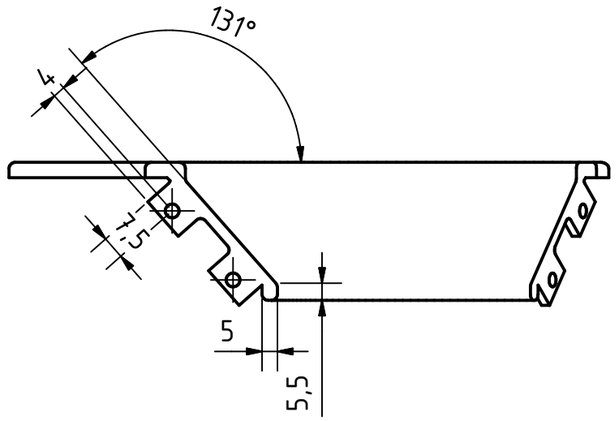
Teilstück Deckel

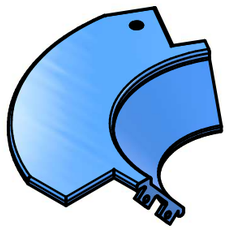
16

A3

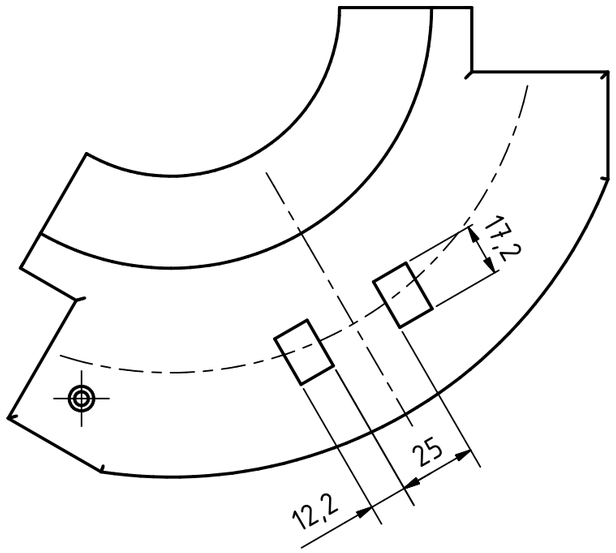
Maßstab: 1:2

Konstrukteur: Simon Bernarding



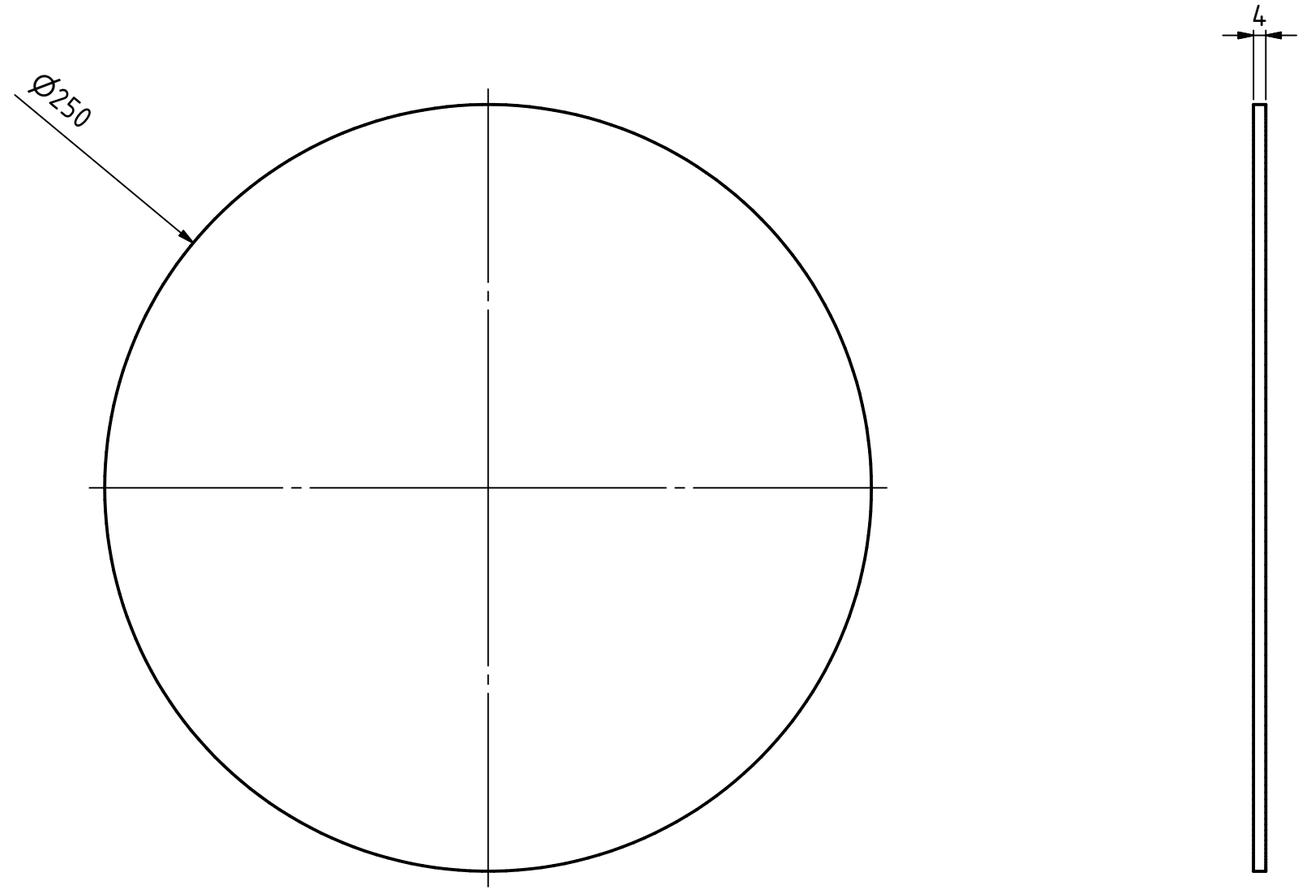


ingenieur wissenschaften htw saar						
				Datum		
				Gezeich	25.03.2026	Industrielle Robotik Ball Ballancer
				Pos. Nr. 14	Teilstück Deckel	17
						A3
Status	Änderungen	Datum	Name	Maßstab: 1:2	Konstrukteur:	Simon Bernarding



Teil Identisch mit Pos. 13, bis auf Schalteraussparungen.

ingenieur wissenschaften htw saar					
				Datum	
				Gezeich	25.03.2026
					Industrielle Robotik Ball Ballancer
				Pos. Nr. 15	Teilstück Deckel Schalter — 18 — A3
Status	Änderungen	Datum	Name	Maßstab: 1:2	Konstrukteur: Simon Bernarding



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					Datum
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Maßstab: 1:2	

Industrielle Robotik
Ball Ballancer

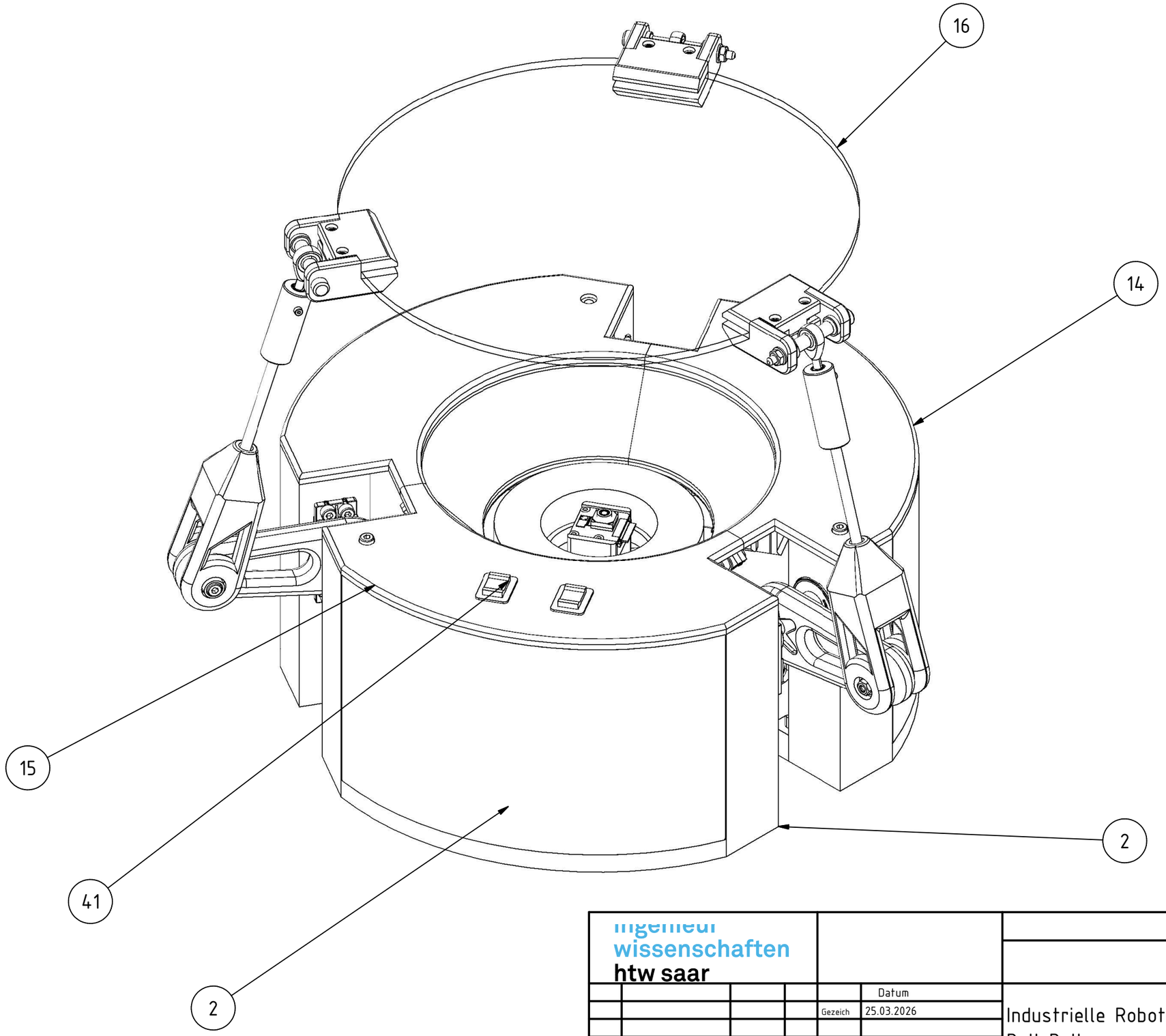
Pos. Nr. 16

Platte

19

A3

Konstrukteur: Simon Bernarding



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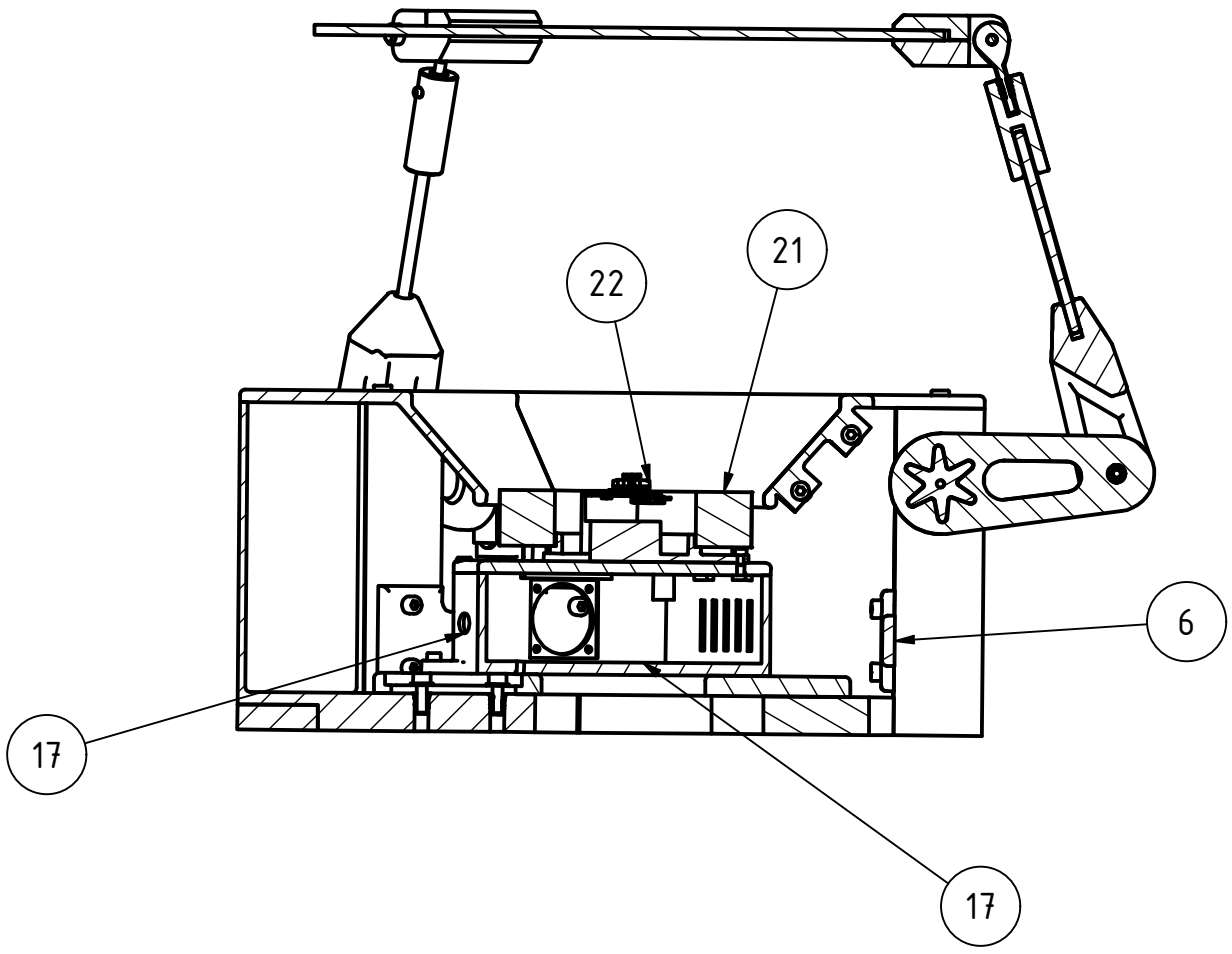
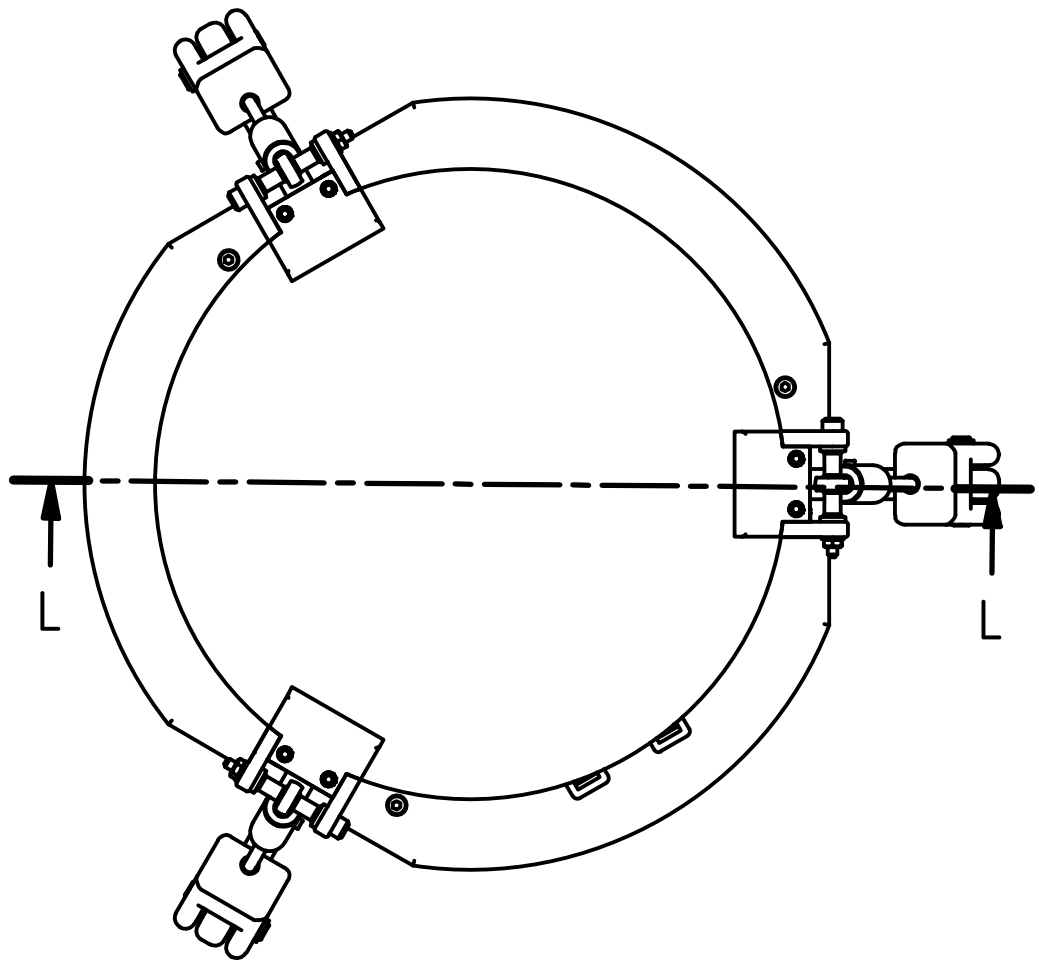
				Datum		
				Gezeich	25.03.2026	Industrielle Robotik Ball Ballancer
				Pos. Nr.		Gesamte Baugruppe — 20 — A3
Status	Änderungen	Datum	Name	Maßstab: 1:2		Konstrukteur: Simon Bernarding

L-L (1 : 3)





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					Datum
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Pos. Nr.	
				Maßstab: 1:2	

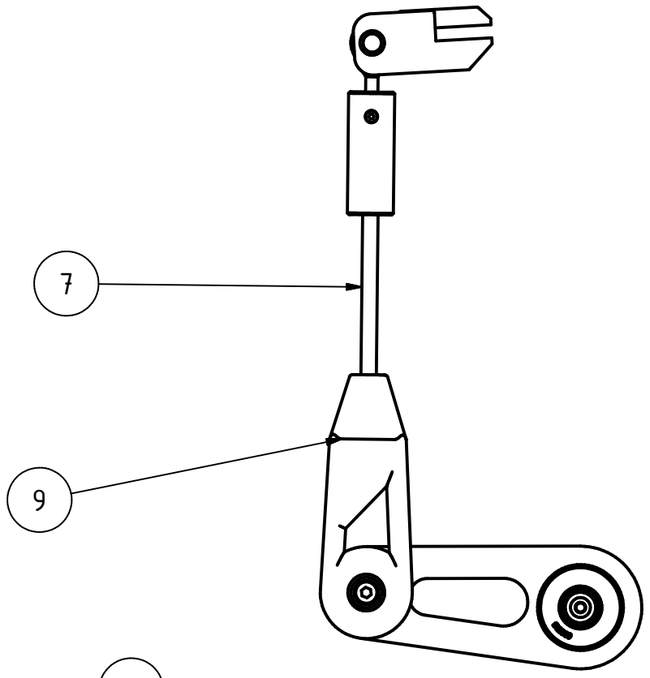
Industrielle Robotik
Ball Ballancer

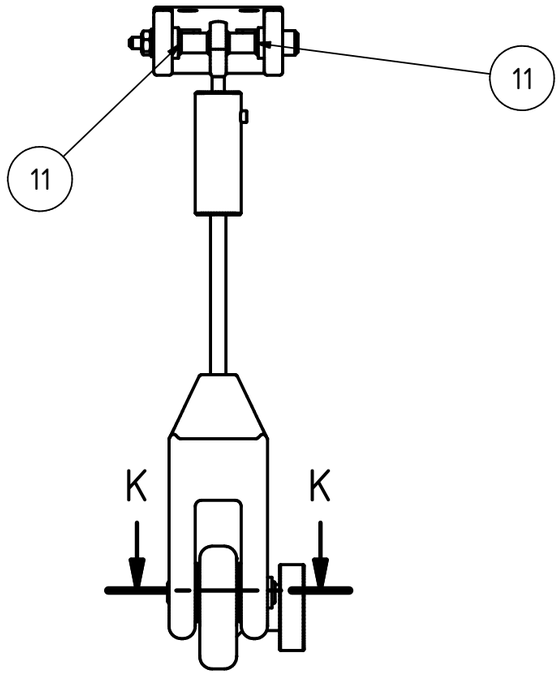
Gesamte Baugruppe

21

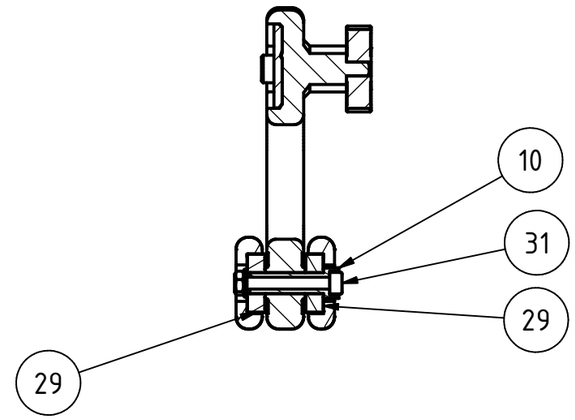
A3

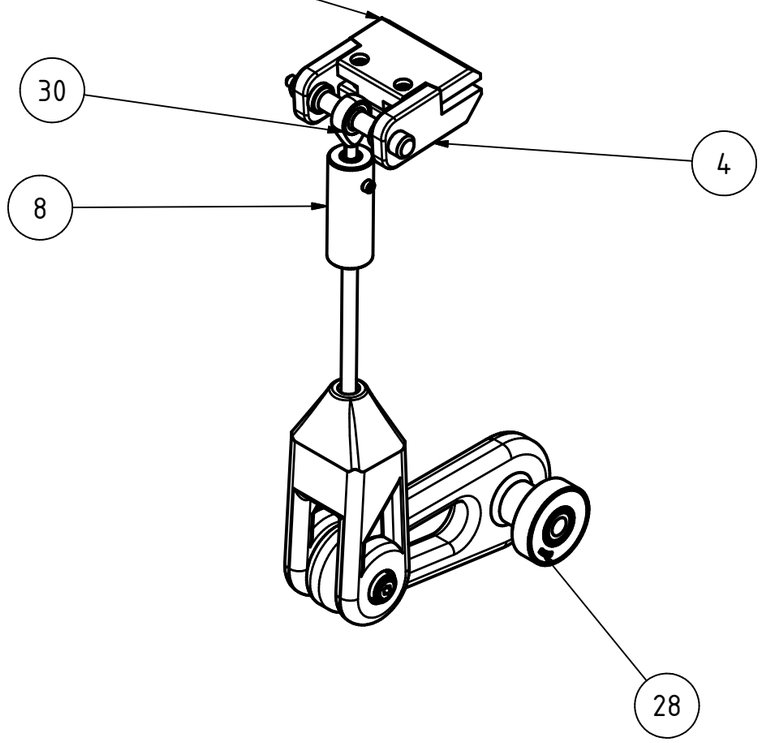
Konstrukteur: Simon Bernarding





K-K (1 : 2)





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				Datum	
				Gezeich	25.03.2026
Status	Änderungen	Datum	Name	Maßstab: 1:2	

Industrielle Robotik
Ball Ballancer

Pos. Nr.

Knickarm Baugruppe

22

A3

Konstrukteur: Simon Bernarding

Stückliste				
POS. NR.	ANZAHL	BZEICHNUNG	BESCHREIBUNG	MATERIAL
1	3	Grundhalter	-	PLA Schwarz
2	3	Abdeckung	-	PLA Schwarz
3	1	Innere Grundplatte	-	PLA Schwarz
4	3	Klemme Unten	-	PLA Weiß
5	3	Klemme Oben	-	PLA Weiß
6	3	Verspannung Lagerturm	-	PLA Schwarz
7	3	Verbinder Schwinge	D = 5 mm	Aluminium
8	3	Schwinge Oben	-	PLA Weiß
9	3	Schwinge Unten	-	PLA Weiß
10	3	Lager Bolzen Knickarm	-	PLA Weiß
11	6	Abstandhalter Klemme	-	PLA Weiß
12	3	Servo Kurbel	-	PLA Weiß
13	3	Bolzen Klemme	-	42CrMo4
14	2	Teilstück Deckel	-	PLA Schwarz
15	1	Teilstück Deckel Schalter	-	PLA Schwarz
16	1	Platte	-	PMMA
17	1	Halter Elektronik Stack	-	PLA Weiß
18	1	Deckel Halter	-	PLA Weiß
19	1	Halter Kamera	-	PLA Weiß
20	1	Halter Ringleuchte	-	PLA Weiß
21	1	LED Ringleuchte	-	-
22	1	Kamera Modul	Camera Moudule 3 v8	-
23	3	Servoarm Sternförmig	-	-
24	3	Amewi AMX Racing Digital Servo	6208MG	-
25	54	Gewindeeinsatz	RX-M4x8	CuZn37
26	3	Gewindeeinsatz	RX-M2x4	CuZn38
27	4	Gewindeeinsatz	RX-M2.5x5	CuZn39
28	3	Rillenkugellager	6001 RS	100Cr6
29	6	Rillenkugellager	607 Z	100Cr6
30	3	Gelenkkopf GT Außengewinde M4	DIN ISO 12240-4 Maßreihe K 63290400	100Cr7
31	3	Zylinderkopfschraube	DIN 912 - M4 x 30	42CrMo4
32	12	Zylinderkopfschraube	DIN 912 - M4 x 12	42CrMo4
33	24	Zylinderkopfschraube	DIN 912 - M4 x 10	42CrMo4
34	12	Zylinderkopfschraube	DIN 912 - M4 x 8	42CrMo4
35	3	Zylinderkopfschraube	DIN 912 - M4 x 6	42CrMo4
36	12	Zylinderkopfschraube	DIN 912 - M3 x 8	42CrMo4
37	4	Zylinderkopfschraube	DIN 912 - M2,5 x 4	42CrMo4
38	3	Zylinderkopfschraube	DIN 912 - M2 x 4	42CrMo4
39	24	Unterlegscheibe	DIN 125 - A 4,3	42CrMo4
40	12	Sechskantmutter	DIN 555-5 - M4	42CrMo4
41	2	Kipp Schalter	12 x 17 mm 12 V	-

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Status	Änderungen	Datum	Name

Datum
Gezeich 25.03.2026

Industrielle Robotik
Ball Ballancer

Pos. Nr. 23
A3

Maßstab:
Konstrukteur: Simon Bernarding